Linear Regression

In a Linear Regression problem, we assume that a set of data points can be modeled accurately by a linear function. We further assume that the observed data points contain some unavoidable, random, noise. In these notes we derive the formula for a best fit line under these assumptions.

Given n points $(x_1, y_1) \ldots (x_n, y_n)$ and an assumed relation $y = f(x) + \epsilon, \epsilon \sim N(\mu, \sigma)$ we want to find a model $f(x_i) = ax_i + b$ such that the root mean squared error

$$\text{RMSE}(a, b) = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\tilde{y}_i - y_i)^2}$$

is minimized. Here we define $\tilde{y}_i = f(x_i)$ and $(\tilde{y}_i - y_i)$ is the *residual* or the error between the observed y_i value and the estimate $\tilde{y}_i$.

First note that if we square the RMSE, the location of the minimum value does not change. Similarly, we can ignore the $\frac{1}{n}$ factor because it does not affect the minimum either. So we can now worry about optimizing the **Residual Sum of Squares** instead.

$$\text{RSS}(a, b) = \sum_{i=1}^{n} (\tilde{y}_i - y_i)^2$$

is minimized.

RSS is a function of the line parameters a and b only. To minimize it we need to take two partial derivatives and then set both partial derivatives to zero. (This could technically find a maximum – but it's reasonably clear this function has no maximum value because the error can always be increased.)

Take partial derivatives

$$
\begin{aligned}
\frac{\partial RSS}{\partial a} &= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) \frac{\partial}{\partial a} (\tilde{y}_i - y_i) \\
&= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) \frac{\partial}{\partial a} (ax_i + b - y_i) \\
&= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) (x_i)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial RSS}{\partial b} &= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) \frac{\partial}{\partial b} (\tilde{y}_i - y_i) \\
&= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) \frac{\partial}{\partial b} (ax_i + b - y_i) \\
&= 2\sum_{i=1}^{n} (\tilde{y}_i - y_i) (1)
\end{aligned}
$$

And now we set both of the partials equal to zero.

$$
\left\{
\begin{array}{l}
\dfrac{\partial RSS}{\partial a} = 0 \\[2ex]
\dfrac{\partial RSS}{\partial b} = 0
\end{array}
\right.
\Rightarrow
\left\{
\begin{array}{l}
\sum_{i=1}^{n} (\tilde{y}_i - y_i) x_i = 0 \\[1ex]
\sum_{i=1}^{n} (\tilde{y}_i - y_i) = 0
\end{array}
\right.
$$

Since $\tilde{y}_i = ax_i + b$

$$
\sum_{i=1}^{n} (ax_i + b - y_i) x_i = 0 \Rightarrow a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i
$$

and

$$
\sum_{i=1}^{n} (ax_i + b - y_i) = 0 \Rightarrow a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} 1 = \sum_{i=1}^{n} y_i
$$

First note that $\sum_{i=1}^{n} 1 = n$. This is simply a linear system of equations in two unknowns and the coefficients are these messy-looking sums. But they're just numbers – coefficients. We can solve this like any linear system.

The most straightforward way to solve any 2x2 system is using Cramer's rule.

$$a = \frac{\begin{vmatrix} \sum_{i=1}^{n} x_i y_i & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} y_i & n \end{vmatrix}}{\begin{vmatrix} \sum_{i=1}^{n} x_i^2 & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & n \end{vmatrix}}$$

and solving for b

$$b = \frac{\begin{vmatrix} \sum_{i=1}^{n} x_i^2 & \sum_{i=1}^{n} x_i y_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} y_i \end{vmatrix}}{\begin{vmatrix} \sum_{i=1}^{n} x_i^2 & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & n \end{vmatrix}}$$

This may look less foreboding if we replace all the sigmas with some new variable names. Let

$$S_x = \sum_{i=1}^{n} x_i, \qquad S_y = \sum_{i=1}^{n} y_i, \qquad S_{xx} = \sum_{i=1}^{n} x_i^2, \qquad S_{xy} = \sum_{i=1}^{n} x_i y_i$$

Taking determinants,

$$a = \frac{nS_{xy} - S_x S_y}{nS_{xx} - S_x^2} \tag{a}$$

$$b = \frac{S_y S_{xx} - S_x S_{xy}}{nS_{xx} - S_x^2}$$

Interpretation as a ratio of variances

Students of statistics may appreciate the following manipulations

Definition of covariance

$$E(xy) - E(x)E(y) = \mathrm{Cov}(x, y)$$

Definition of variance

$$\mathrm{Var}(x) = E\left[(x-\mu)^2\right]$$

Lemma

$$\begin{aligned}
\mathrm{Var}(x) &= E\left[(x-\mu)^2\right] \\
&= E\left(x^2\right) - 2\mu E[x] + E[\mu]^2 \\
&= E\left[x^2\right] - 2E[x]^2 + \mu^2 \\
&= E\left[x^2\right] - E[x]^2
\end{aligned}$$

Manipulating the denominator of equation (a) above,

$$\begin{aligned}
n\sum x_i^2 - \left(\sum x_i\right)^2 &= n^2\left(\frac{1}{n}\sum x_i^2 - \left(\frac{\sum x_i}{n}\right)^2\right) \\
&= n^2\left(E\left[x^2\right] - E[x]^2\right) \\
&= n^2\,\mathrm{Var}(x)
\end{aligned}$$

And the numerator

$$\begin{aligned}
&n\sum x_i y_i - \sum x_i \sum y_i \\
&\qquad = n^2\left(\frac{1}{n}\sum x_i y_i - \frac{1}{n}\sum x_i \cdot \frac{1}{n}\sum y_i\right) \\
&\qquad = n^2\left(E\left[xy\right] - E[x]E[y]\right) \\
&\qquad = n^2\left(E[xy] - \mu_x\mu_y\right) \\
&\qquad = n^2\,\mathrm{Cov}(x,y)
\end{aligned}$$

so

$$a = \frac{E[xy] - \mu_x\mu_y}{E\left[x^2\right] - \mu_x^2} = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}$$